

USART / UART

Presentation

Group 10

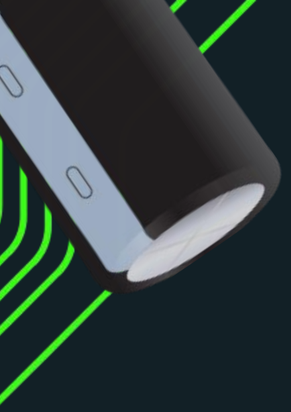
Our Team

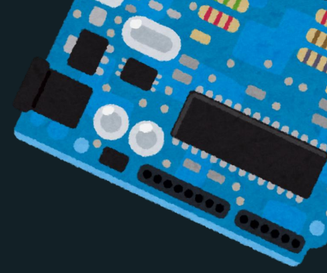
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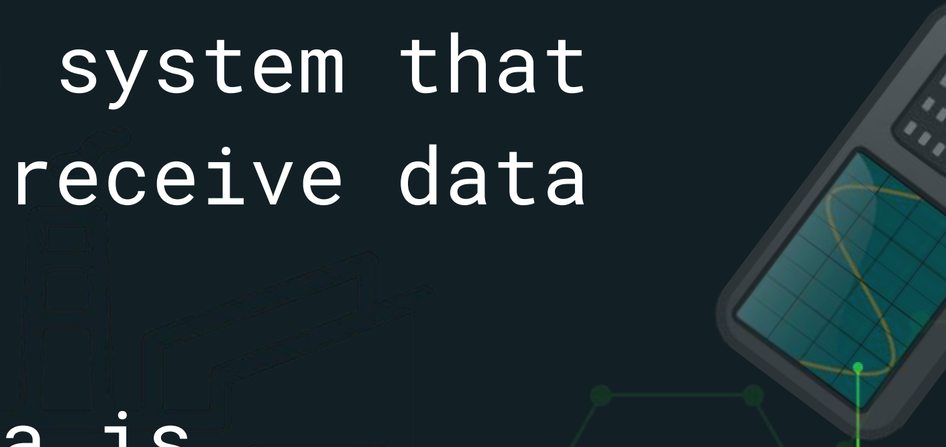


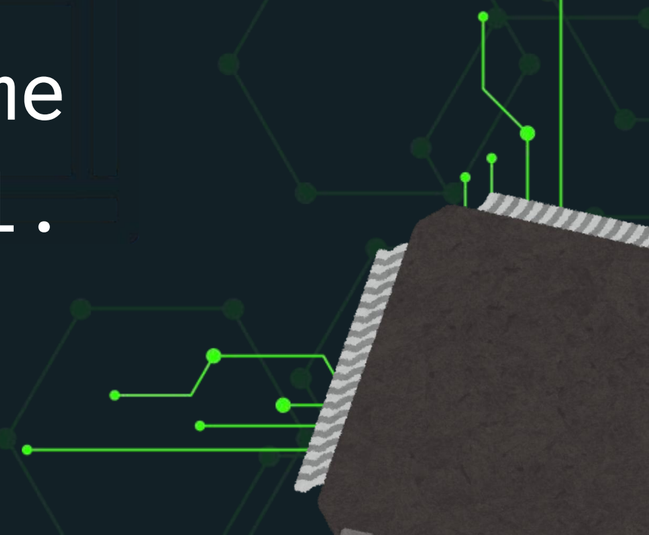


What is UART / USART ?

UART/USART is a communication system that allows the STM32 to send and receive data serially.

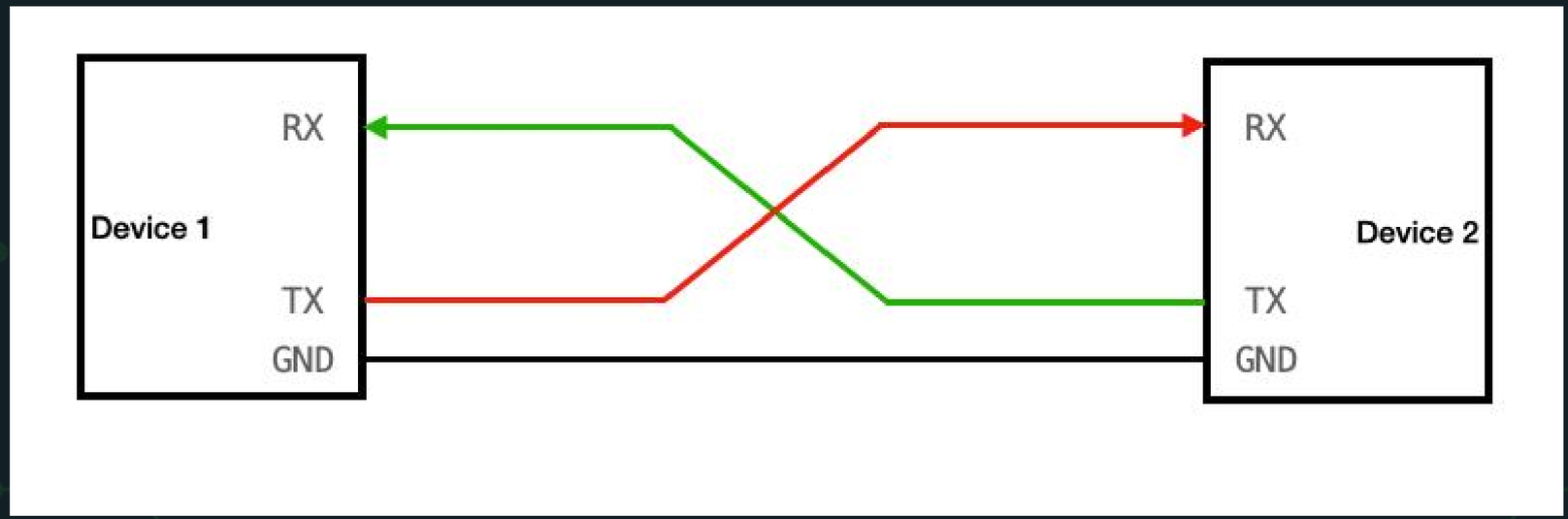
Serially means data is transmitted one bit at a time over a communication channel.





UART = Universal Asynchronous Receiver/Transmitter

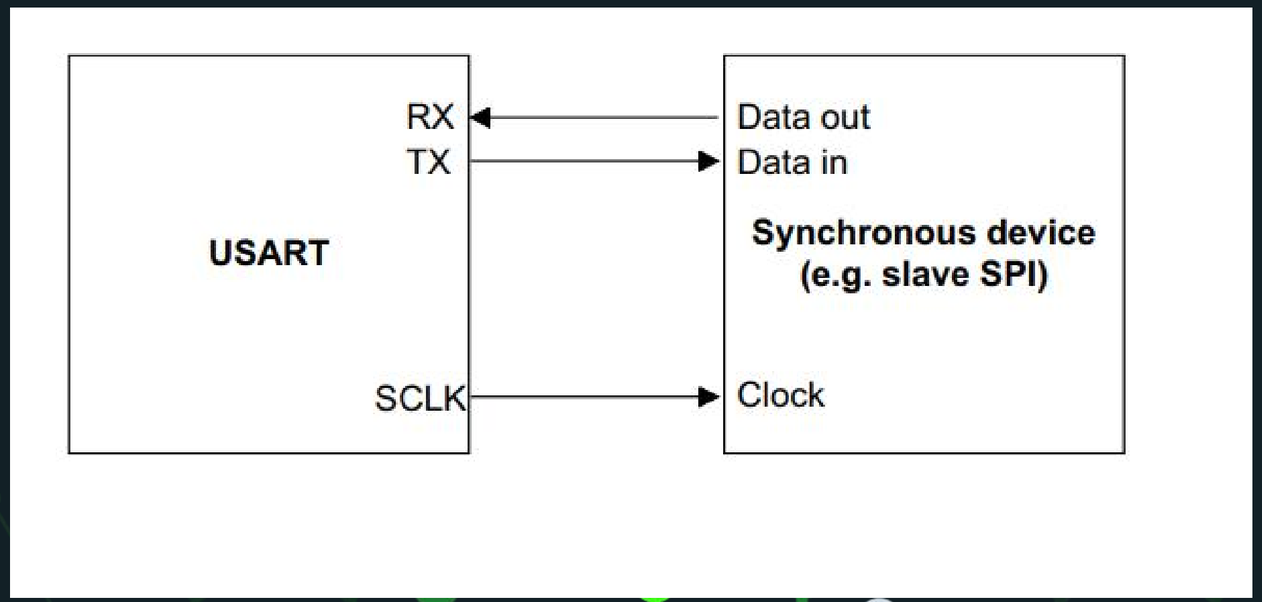
UART is a hardware peripheral/protocol used to transmit and receive serial data without a separate clock signal.



USART = Universal Synchronous/ Asynchronous Receiver/Transmitter

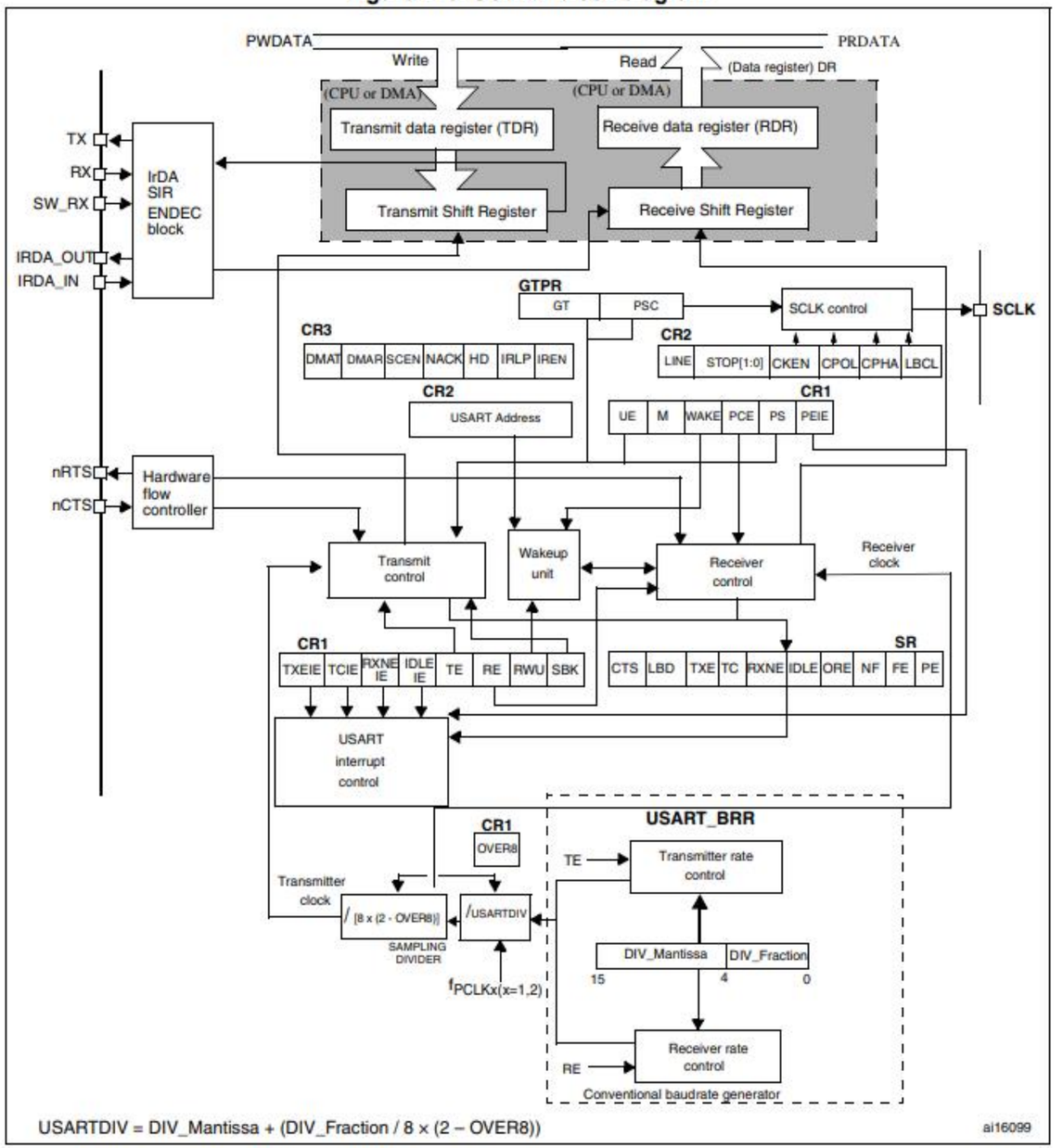
USART can do UART-style asynchronous communication, plus synchronous communication.

This is why the peripheral is called USART



USART Block Diagram

Figure 278. USART block diagram



What are TX and RX?

TX = Transmit
It sends data.

RX = Receive
It receives data.

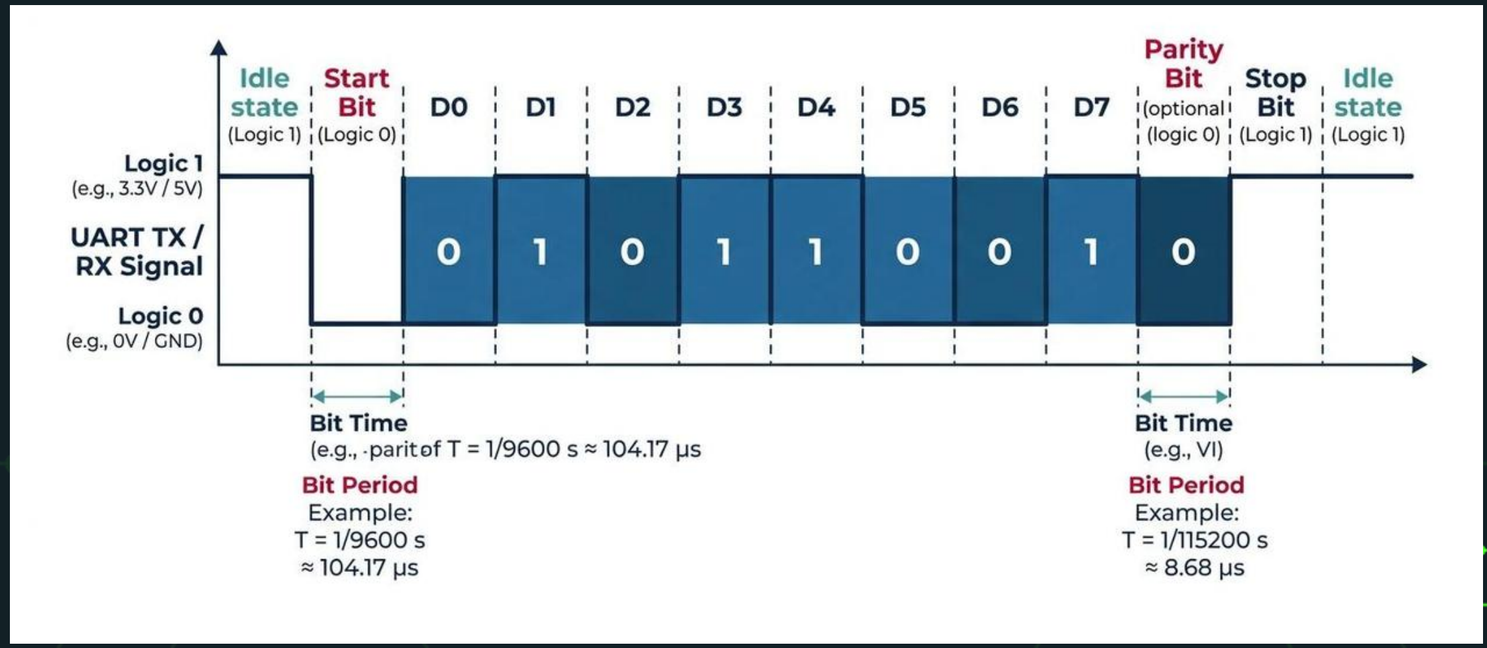
Why do we need GND?

We need GND (ground) in UART because voltage is a relative measurement. Without a shared reference point, the receiving device cannot distinguish between high logic (e.g., 3.3V or 5V) and low logic (0V)

UART Frame Structure

A UART frame consists of:

- Start Bit
- Stop Bit
- Data Bits
- Parity Bit(optional)



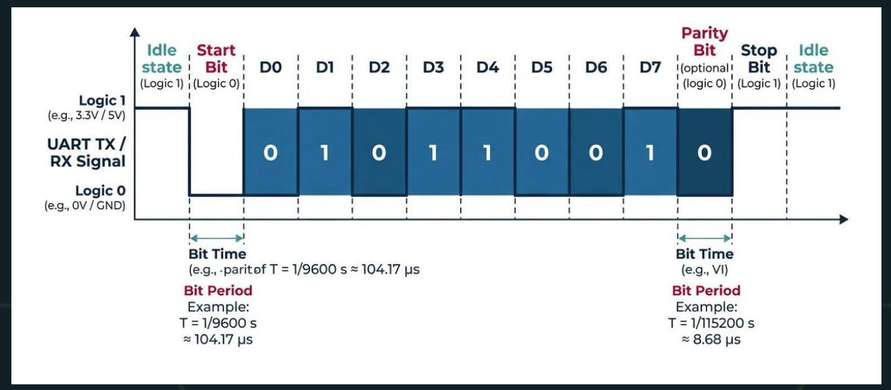
Start Bit:
Tells the receiver a new frame is starting

Data Bits:
Usually 5–9 bits
but most common is 8 bit

Parity:
Used for simple error detection.

- Even parity
- Odd parity
- No parity

Stop Bit:
Indicates the end of the frame.
1 stop bit or 2 stop bits



Example: Sending the Character 'A'

ASCII: A = 65

Binary:01000001

For an 8N1 UART frame:

START	DATA	STOP
0	01000001	1
|	|	|
Start	8 bits	Stop

UART normally transmits the data bits LSB first.

1 0 0 0 0 0 1 0

Baud Rate

Baud rate represents the number of symbols transmitted per second

Common values:

- 4800
- 9600
- 19200
- 38400
- 57600
- 115200

Example:Baud Rate = 9600

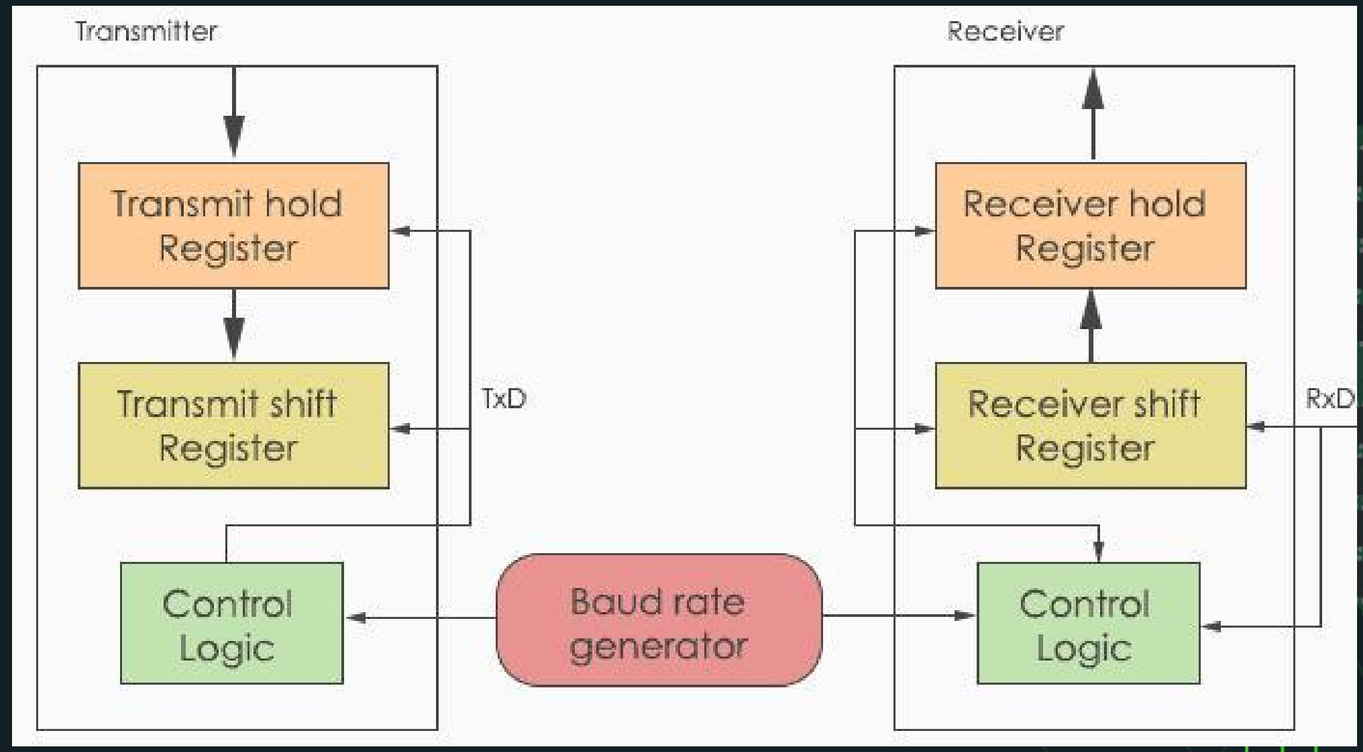
So approximately 1 bit = 1/9600

= 104.17 µs

USART Data Transmission and Reception

For transmission, the CPU puts data into the Transmit Data Register. The USART moves the data to the transmit shift register, converts it into a serial frame, and sends it through the TX pin.

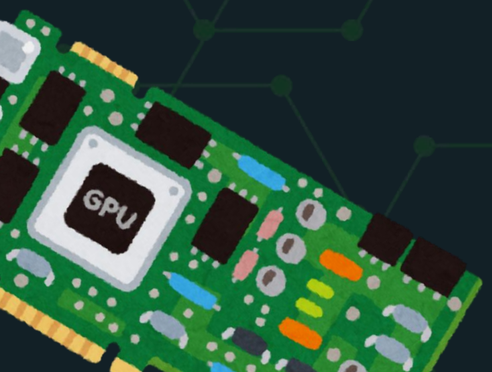
For reception, serial data arrives through the RX pin. The USART receives and reconstructs the frame, stores the data in the Receive Data Register, and the CPU can then read it.

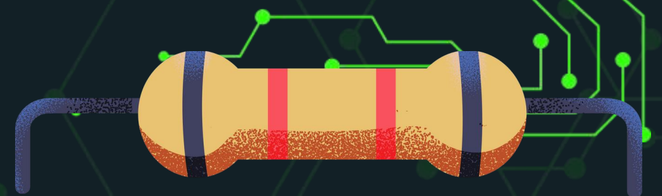


USART Registers

Register	Purpose
USART_SR	Status information
USART_DR	Data transmission/reception
USART_BRR	Baud-rate configuration
USART_CR1	Main control
USART_CR2	Additional control
USART_CR3	DMA/flow-control/other features
USART_GTPR	Guard time & prescaler












Thank
You